

Digital Image Processing Assignment

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Q.1: Discuss in detail about the Smoothing frequency domain filter?

Ans: Introduction: $\rightarrow$ Smoothing in the frequency domain is achieved by suppressing or attenuating high-frequency components of an image's Fourier Transform. Since high frequencies correspond to sharp transitions, edges, and noise, removing them produces a smoother (blurred) result.

The general filtering model is:

$$G(u,v) = H(u,v) \cdot F(u,v)$$

where $F(u,v)$ is the Fourier Transform of the input image, $H(u,v)$ is the filter transfer function, and $G(u,v)$ is the filtered output.

The steps involved in frequency domain filtering are:

1. multiply the input image by $(-1)^{x+y}$ to center the transform.
2. compute $F(u,v)$, the DFT of the image.
3. multiply $F(u,v)$ by the filter function $H(u,v)$.
4. compute the inverse DFT of the result.
5. obtain the real part of the result.
6. multiply again by $(-1)^{x+y}$.

There are three major types of smoothing (low-pass) filters:

1. Ideal low pass filter (ILPF):

The transfer function is defined as:

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$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{if } D(u, v) > D_0 \end{cases}$$

where D_0 is a positive constant called the cutoff frequency and $D(u, v)$ is the distance from a point (u, v) to the center of frequency rectangle.

$$D(u, v) = \left[\left(u - \frac{M}{2} \right)^2 + \left(v - \frac{N}{2} \right)^2 \right]^{1/2}$$

Key properties $\rightarrow$

- All frequency on or inside a circle of radius D_0 are passed with -out any attenuation.
- All frequencies outside the circle are completely blocked.
- The filter is radially symmetric, it is completely defined by its radial cross section.
- The transition b/w $H=1$ and $H=0$ is instantaneous (sharp cut)

Drawback $\rightarrow$ The abrupt cutoff causes ringing artifacts in the spatial domain output, making the ILPF impractical for many apps

2. Butterworth low pass filter (BLPF) $\rightarrow$

The transfer function of a Butterworth LPF of order n with cutoff frequency D_0 is

$$H(u, v) = \frac{1}{1 + \left[\frac{D(u, v)}{D_0} \right]^{2n}}$$

Key properties $\rightarrow$

- unlike the ILPF, the BLPF does not have a sharp discontinuity. There is a smooth, gradual transition b/w passed & filtered frequency.
- For $n=1$ the filter is very smooth, as n increases, the filter approaches the behaviour of an ILPF.
- At $D(u, v) = D_0$, $H = 0.5$ for all orders.
- Higher order filter produce more filtering but also more ringing.
- It offer a good balance b/w smoothing and artifact reduction.

3. Gaussian low pass filter (GLPF) $\rightarrow$

The transfer function is define as $\rightarrow$

$$H(u, v) = e^{-D_0^2 D(u, v)^2 / 2}$$

where, $D(u, v)$ is the distance from the origin of the Fourier transform and D_0 is the cutoff frequency.

Key properties $\rightarrow$

- The GLPF has the smoothest possible roll-off - no ringing artifacts in the spatial domain.
- It is the most blurred filter when applied to an image.

Q.2 $\rightarrow$ What is Inverse filtering? Explain how inverse filtering restores blurred images?

Ans $\rightarrow$ Inverse filtering is an image restoration technique used to recover original image from blurred image using knowledge of degradation function.

Degradation model $\rightarrow$

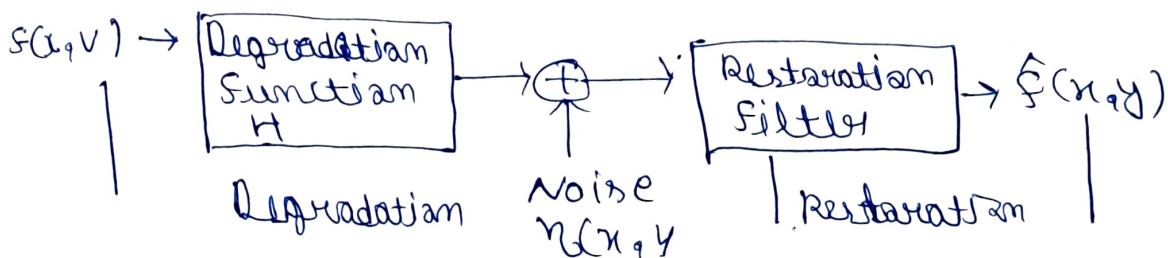
$$G(u, v) = \underset{\substack{\uparrow \\ \text{degraded image}}}{H(u, v)} \underset{\substack{\uparrow \\ \text{original image}}}{F(u, v)} + \underset{\substack{\uparrow \\ \text{noise}}}{N(u, v)}$$

Inverse filter $\rightarrow$

$$S(u, v) = \frac{G(u, v)}{H(u, v)}$$

(*) Working of Inverse filtering $\rightarrow$

- $\rightarrow$ Convert blurred image to frequency domain using Fourier transform.
- $\rightarrow$ Divide degraded image by degradation function
- $\rightarrow$ Apply Inverse Fourier transform
- $\rightarrow$ Obtain restored image.



Limitation $\rightarrow$

- Very sensitive to noise
- cannot restore image if $H(u, v) = 0$
- Amplifies noise at high frequency.

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Q.3 $\Rightarrow$ Define Image segmentation what is meant by region based Segmentation?

Ans $\Rightarrow$ Image segmentation is the process of dividing an image into multiple meaningful regions for easier analysis.

Goal $\Rightarrow$

- Simplify image representation.
- ~~Learn~~ Identify objects.
- Detect Boundaries.

Types of Segmentation $\Rightarrow$

1. Threshold based
2. Edge based
3. Region based
4. Clustering based

Region based segmentation $\Rightarrow$

$\rightarrow$ Region based segmentation groups pixel based on similar properties such as $\Rightarrow$

$\rightarrow$ intensity
 $\rightarrow$ Color
 $\rightarrow$ texture.

Pixel is same region label $\Rightarrow$

$R_i = \text{True}$.

Region satisfies homogeneity condition.

Types of regional based methods $\Rightarrow$

Q. Region Growing $\Rightarrow$

Shape $\Rightarrow$ Select seed pixel

$\rightarrow$ compare neighbouring pixels.

$\rightarrow$ ~~can~~ add similar pixel.

$\rightarrow$ Repeat until remain complete.

- Q.3. Region extracting and ~~merging~~ merging?
- Image divide into smallest region.
 - Merge similar region.
 - Produces continuous ~~the~~ ^{the} regions.
 - more meaningful segmentation. Advantage

Q.4. → Describe different estimation methods of degradation fun. Explain how Wiener filter restores washed image?

Ans: → Image degradation model

$$G(u, v) = H(u, v) f(u, v) + N(u, v)$$

Degradation sources: →

- Motion blur
- Defocus blur
- Atmospheric turbulence
- Sensor noise.

method to estimate degradation function $H(u, v)$

1. observation method: →

- Analyse ~~the~~ degraded image visually
- Detect ~~degradation~~ blur type.

2. Experiment model: →

$$\text{Motion blur: } H(u, v) = \frac{\sin \pi b (\sqrt{u^2 + v^2})}{\pi b (\sqrt{u^2 + v^2})} e^{-j\pi b (u^2 + v^2)}$$

3. Mathematical modelling: →

- capture image with known blur.
- Estimate ~~spectrum~~ ^{spectrum} and ~~restored~~ ^{restored} degradation mathematically.

4. To use spectrum method: →

- Compare power spectrum of degraded image with original.

Wiener filter:

Linear filter is optimal restoration filter that minimizes mean square error b/w original and restored image.

$$\hat{F}(u, v) = \frac{H(u, v)}{|H(u, v)|^2 + S_n(u, v)} G(u, v)$$

$\nwarrow$ Noise Power Spectrum $\nearrow$ Original Image Power Spectrum

⊗ Working of linear filter:

- Estimate degradation function.
- Estimate noise.
- Apply linear formula
- Obtain restored image

⊗ Advantages:

- work even with noise
- Produces optimal result
- minimize error.

